

AAA are rated at 1.5 volts, so select a four-volt range on the DC scale for such batteries. For testing a nine-volt battery, move the selector to 40-volt range.

Touch each end of the battery with a probe. If red and black wires are reversed, a minus (-) sign will appear in front of the numbers. A 1.5-volt battery is dead if it registers one volt or less. If a device does not work but the 1.5-volt battery registers, for example, 1.38 volts, then the issue is with the device. A nine-volt battery is dead when it fails to read above 7.5 to eight volts.

When lamps leave you in the dark

There are two possibilities if a bulb does not glow; either the light bulb has burnt out or there is a problem with the lamp. This can be clarified by using a multimeter.

First, set the multimeter to measure resistance in ohms. If there is more than one resistance setting, choose the most sensitive one. Set the meter to 400-volt AC selection. Touch one probe to the side of the bulb socket, and the other to the tip at the bottom of the socket. The meter should read about 230 volts.

Sometimes, expected voltage in the bulb socket and the bulb is good, but the lamp still does not light up. In that case, the problem is with the socket.

Note. This test will work only for an incandescent light bulb.

To check the fuse

From household gadgets to automobiles almost all electronic instruments use fuses. If such devices are not working, first thing to do is to check the fuse. To do so, turn off the device, disconnect the power and remove the fuse. Set the multimeter to CONT. (continuity) and do the continuity test by putting the positive and negative leads on either side of the fuse and check the reading. If the meter beeps there is continuity



Fig. 3: Checking a fuse



Fig. 4: Checking the capacitor

and the fuse is alright, but if there is no sound, it means the fuse has gone bad.

When the air-conditioner quits the job

It might be during the worst weather conditions that your air-conditioner fails to work. Some multimeter tips and tricks could help you find out what actually went wrong, whether you can fix it yourself or you need to call an electrician.

Initial checks. There is a set of circuit breakers in your main electrical panel. Check to see if the circuit breaker switch has moved to off position due to a sudden overload. If it is still in on position and yet the air-conditioner is not working, shut off all power to the air-conditioner.

Check the fuses for continuity. If the fuses are fine, discharge the air-conditioner's condenser (capacitor) by putting the metal tip of a screwdriver between the two terminals.

Checking the condenser. Disconnect the wires of the condenser and set your multimeter to read ohms or resistance. Place the black lead in the hole marked com, and the red lead into the hole that says cap or has ohms icon by it. Place one lead on each of the condenser's terminals and check the reading you get.

This reading should be within six per cent of your particular capacitor's rating. If it is not, your capacitor is faulty and must be replaced.

To fix laptops

Power supply is a critical component in your computer, and when this begins to fail, your computer may switch off unexpectedly, experience start-up warnings or over-heat. Several simple checks with a digital multimeter can uncover some of the warning signs of the impending hardware failure.

Precautions. Unplug your computer and disconnect all cables connected to it. If your unit is equipped with a power switch, turn the power supply off. Place your computer on an anti-static mat. Remove the outer casing and unplug all power connectors. See the sticker on the power supply and check voltage information.

Checking the power supply. Turn the power supply on and switch your multimeter to the range that is appropriate for reading the voltage from your power supply. Locate the largest group of wires coming out of the power supply. Connect the multimeter's negative (black) probe to the ground-wired pin, which will be black in colour. Connect the positive (red) probe to the power line, which will be green in colour.

Record the voltage that appears on your multimeter. If your power supply sticker shows a voltage of 135 volts, you should be seeing the same reading on the multimeter. If you see a much lower or higher reading, your power supply has malfunctioned and needs to be replaced.

Differentiate a hot and neutral wire

Use a pair of protective rubber gloves while doing this test. You may accidentally touch a hot wire and get an electric shock. Set your multimeter to measure voltage. Place the multimeter's red probe on the bare metal on the end of one of the wires. The positive probe on a multimeter is red, and the negative one is black. Place the probe of the multimeter's black wire on the bare metal on the